James Jagielski

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https://james-jagielski.github.io/Project-Portfolio/
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EDUCATION

Columbia University | Master of Science | *PhD Candidate under Ken Sheppard* **New York, NY**
- **Clubs**: Sailing and Table Tennis. **Relevant Coursework**: VLSI Design Lab, Advanced Analog IC, mmWave IC, microwave circuit design, Analog-Digital Interfaces, Power Management IC, Harsh Environment electronics, Digital VLSI Fall 2026

Olin College of Engineering | Bachelor of Science: Electrical and Computer Engineering **Needham, MA**
- Sailing team Captain. **Relevant Coursework**: Mixed Analog-Digital VLSI, Microelectronic Circuits, Software Design, Computer Architecture, Advanced Computer Architecture, Power Electronics, Sensors & Instrumentation May 2025 GPA: 3.85/4.0

Southwest High School **Minneapolis, MN**
- IB Diploma; Varsity team captain of swimming, sailing and table tennis; Jazz alto saxophonist June 2021 | GPA: 3.95/4.0

EXPERIENCE

Columbia University **New York, NY**
Bioelectronic Systems Laboratory | Ken Shepard Spring 2026
- 60 GHz receiver in 90nm with LNA and MIXER. VCO has 7.3% tuning range and phase noise of -94dBc/Hz at 1MHz offset. The receiver has NF of 4.28dB, IP1dB of -29.5dB, power of 53.5mW, gain of 30dB up to 3GHz. LNA S11 is -18.2dB at 50 Ω.
- 12GHz 2-stage LNA: NF of .5dB, gain of 21dB, output reflection of -27dB, input reflection of -15dB, and unconditionally stable.
- An 8-bit 250-MSample/s Current-Steering DAC with a range of 1.4V, SFDR of 52.75 dB, and INL & DNL lower than +/- .5LSB.
- Designed a Switched-Cap fully differential amplifier for 5µW and .7V VDD rail in a sensing and stimulation device for dry eye disease, placed in the tear duct. This circuit will be taped out in TSMC 130nm, with PCB test board, and will be tested Fall 2026.
- A 1 MHz Buck Converter with Type-III Compensation. 3V-1V conversion, efficiency of 92%, output is 3W with 3A load current.
- Closed-Loop Dynamic Biasing for Threshold Voltage Stability in harsh environment disturbances in LEO Beamforming Systems.

Course Assistant | Digital VLSI Circuits Fall 2025
- Supported students in developing digital VLSI design in TSMC 65nm and finFET N16 TSMC technology with a mini processor.

Course Assistant | Digital Systems Laboratory Spring 2026
- Basic lab course in the design of digital systems. Supported students in legacy LSI logic families (such as TTL) as well as data conversion, timing circuits, data communications, transmission lines, and programmable logic on bread board circuitry.

Semtech Corporation **San Diego, CA**
Test Engineering Intern Summer 2024
- Wrote test programs for post-silicon testing on Advantest V93000 and Teradyne flex and testing communication protocols

Olin College **Needham, MA**
MADVLSI Circuits & Systems Lab | Dr. Bradley Minch Summer and Fall 2023
- Designed a source measuring unit on a 4-layer PCB (to characterize semiconductor devices) for use in the microelectronics course at Olin, wrote the associated firmware, and made a MATLAB GUI for easier use

Olin Plasma Engineering Lab | Electrical Sub Team Lead Fall 2023
- Worked on the high-power circuitry for the hall thruster and set up electrical circuitry in plasma labs at MIT and Northeastern
- Designed a testing rig for the hall thruster to measure the force and temperature and collected the data for physics analysis
- Designed a 2.45GHz and 30W electrical system into a molybdenum antenna for an Electron Cyclotron Resonance Thruster

Course Assistant | Introduction to Microelectronic Circuits Spring 2024
- Supported students in learning transistor-level design of current mirrors, differential pairs, and single-stage op-amps

Course Assistant | Introduction to Sensors, Measurement, and Instrumentation Spring 2023
- Counseled students in designing analog signal processing circuits with an emphasis on op-amps, and passive and active filters

SKILLS

- **Electrical Design and Prototyping:** Virtuoso, ADS Momentum, Magic (VLSI Layout), DRC/LVS, postlayout simulations, xschem, NGSpice, LTSpice, KiCAD, Altium, Monte Carlo, floorplanning, power analysis, block diagrams, design reviews
- **Electrical Instrumentation:** Oscilloscope, Function Generator, Source Measure Unit, Multimeter, Current Clamp
- **Software and programming:** MATLAB, Python, C, Verilog, GTKwave, COMSOL, SOLIDWORKS
- **IC Circuit Design:** Bandgap reference, op-amps, comparators, biasing circuitry, analog-to-digital converters, digital-to-analog converters, current mirrors, switched-capacitor circuits, digital logic cells, low-power moderate inversion CMOS design

PROJECTS

A Bit-Serial Algorithmic Bi-Directional A/D/A Converter in SKY130 Process **Mixed Analog-Digital VLSI**
- Designed 8-bit A/D/A converter and explored floating gate differential pairs for the OTAs and comparator

Self-Biased Two-Stage Cascode Op-Amp in SKY130 Process **Mixed Analog-Digital VLSI**
- Converted an op-amp design from 5V process to a 1.8V process using low voltage current mirrors

• Optimized matching using proper analog layout techniques (common centroid, same surround, and dummy devices)

7-bit M-2M Digital-to-Analog Converter in SKY130 Process **Mixed Analog-Digital VLSI**
• Designed a current bias for the D/A converter using a bootstrap current mirror
• Designed a CSRL Shift Register to insert bits into the D/A converter
• Simulated the differential and integral nonlinearity and Monte Carlo to examine accuracy of the D/A converter

RISC-V Single-Cycle Scalar CPU **Computer Architecture**
• Implemented CPU using structural Verilog, wrote test benches and simulated using GTKwave

Asynchronous SAR ADC in SKY130 Process **Mixed Analog-Digital VLSI**
• Designed an ultra-low power level crossing SAR ADC with handshake based digital control circuitry

MD5 Hashing Accelerator TapeOut in SKY130 Process **AVLSI LAB**
• Designed, laid out, and tested custom standard cells for the Asynchronous VLSI research lab at Yale University